

Technology Stack

Date	15 March 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS	HTML and CSS are used to build a simple, responsive interface for entering soil parameters and displaying crop recommendation results.
2	Backend / Server-Side	Python, Flask	Flask provides the web framework for handling user requests and integrating the machine learning models with the application.
3	Database / Data Storage	CSV Dataset, Pickle (.pkl) Models	The crop dataset is stored as a CSV file, while trained machine learning models are stored as .pkl files for fast prediction.
4	Cloud / Hosting / Deployment	Local Deployment (Flask)	The application runs locally for development and demonstration and can be deployed to cloud platforms in the future.
5	Version Control & CI/CD	Git, GitHub	Git and GitHub are used for version control, collaboration, source code management, and project backup.
6	Third-Party Libraries / Tools	NumPy, Pandas, Scikit-learn, Matplotlib, SciPy	These libraries are used for data processing, visualization, machine learning model development, and scientific computing..